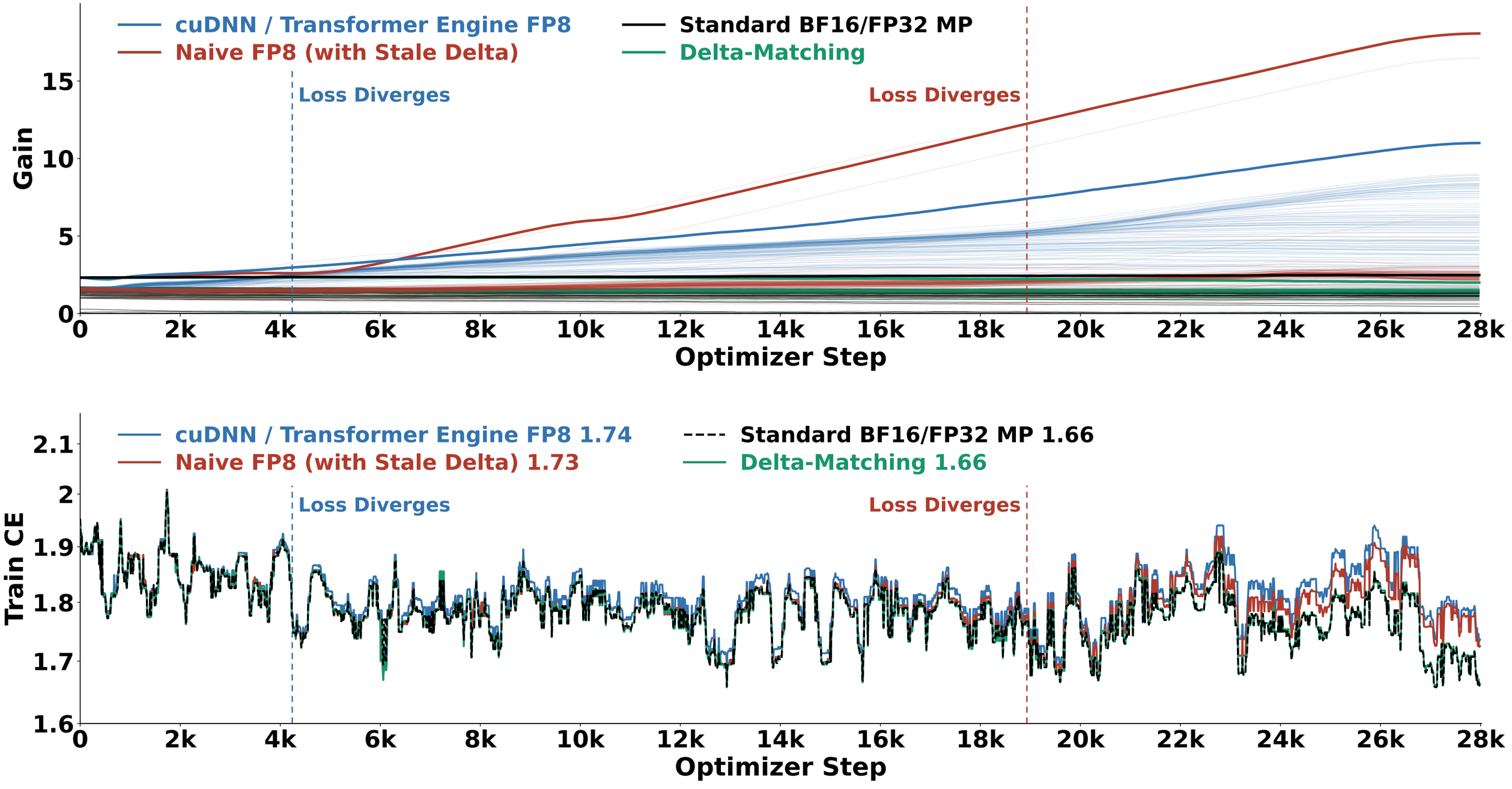

cuDNN / Transformer Engine FP8
Naive FP8 (with Stale Delta)
Standard BF16/FP32 MP
Delta-Matching
Loss Diverges
Loss Diverges
Gain
0
5
10
15
0
2k
4k
6k
8k
10k
12k
14k
16k
18k
20k
22k
24k
26k
28k
Optimizer Step
cuDNN / Transformer Engine FP8 1.74
Naive FP8 (with Stale Delta) 1.73
Standard BF16/FP32 MP 1.66
Delta-Matching 1.66
Loss Diverges
Loss Diverges
Train CE
1.6
1.7
1.8
1.9
2
2.1
0
2k
4k
6k
8k
10k
12k
14k
16k
18k
20k
22k
24k
26k
28k
Optimizer Step